

Vinod A

Data Engineer

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Professional Summary

Proficient Azure Data Engineer with 4+ years of experience specializing in the healthcare and consumer goods company. Expertise in building scalable data pipelines and optimized data warehouses using Azure services and Databricks.

Technical Knowledge

- **Skills:** ETL, Data Warehousing, Data Profiling and validation, Azure, ETL Process optimization
- **Languages & Databases:** MS SQL Server, Oracle, SQL, Python, Pyspark
- **Tools:** Azure Synapse Analytics, Azure data factory, Databricks, Azure ADLS, Azure logic apps, Snowflake, Informatica PowerCenter, Putty
- **Libraries used:** logging, traceback, koalas, json, pandas, os, datetime, csv, delta table, sqlalchemy, concurrent.features, yaml, pyodbc. Pyspark. window, pyspark. functions, requests.
- **Support tools:** JIRA, Azure Devops

Professional Experience

Deloitte, Coimbatore

Consultant – Data Engineer | Apr 2024 - Present

- **Designed and developed a scalable, end-to-end Master Data Ingestion Framework** in Databricks, consolidating all master data sources into a single notebook, reducing notebook count by 80% and minimizing manual intervention and maintenance efforts.
- **Implemented a unified, dynamic merge logic for dimension tables** using an Active Indicator Flag, seamlessly managing inserts and updates across 15+ dimension tables and enabling zero-code onboarding for new data sources.
- **Engineered a high-performance Fact Data Ingestion pipeline** capable of processing millions of records daily, leveraging checkpointing, broadcasting, and other Spark optimization techniques to improve processing efficiency by up to 40% and reduce runtime variability.
- **Developed a dependency-aware retriggering mechanism** in Databricks, automatically detecting upstream master data refreshes and triggering downstream fact pipelines—achieving zero manual intervention.
- **Implemented a comprehensive pipeline execution logging framework**, capturing metadata such as pipeline type, execution status, timestamps, runtime metrics, and error details for enhanced traceability and monitoring.
- **Integrated the logging framework with Azure SQL Database**, providing real-time pipeline status updates to a web application and enabling robust auditing and monitoring capabilities.
- **Utilized Git-based version control (Azure Repos)** to manage ADF and ADB pipeline versions, performing code reviews and pull requests to maintain high-quality, version-controlled deployments.

Cognizant Technology Solution, Coimbatore

Associate- Data Engineer | Jun 2021 – Mar 2024

- **Developed a comprehensive, end-to-end data ingestion pipeline** with multi-configuration capabilities, ingesting data from **10+ diverse sources (files, databases, APIs)** into a refined storage layer, processing 5+ TB of data monthly.

- **Designed a reusable ingestion framework to handle 5+ file formats** (CSV, fixed-length, Excel, Parquet, JSON) with 100+ configurable features, reducing ingestion setup time by 40% and improving framework scalability.
- **Built and maintained Azure Data Factory (ADF) and Synapse pipelines** for incremental and full data loads, enhancing data load efficiency and reliability across multiple databases.
- **Optimized infrastructure and performance by implementing parallel processing** using Python multiprocessing and PySpark, enabling better resource utilization and reducing cloud infrastructure costs.
- **Performed advanced data refinement in PySpark**, including duplicate detection, null handling, schema validation, and DDL enforcement, improving data quality and reducing downstream errors by 50%.
- **Automated workflow orchestration through ADF triggers and parameterized pipelines**, achieving 100% dynamic scheduling coverage and eliminating manual execution dependencies.
- **Implemented pipeline execution logging into Azure SQL Database** for real-time traceability, monitoring, and issue diagnosis, significantly reducing resolution time.
- **Developed automated testing scenarios for ADF** pipelines across data layers, reducing testing time by 70% and minimizing manual validation efforts.

Education

Dr Mahalingam College of Engineering and Technology, Pollachi
Batchelor of Engineering — Electrical and Electronics
Anna University • Chennai • 2021 • 8.75

Certification

- Azure Fundamentals (AZ-900)
- Azure Data Fundamentals (DP-900)
- Azure Data Engineer Associate (DP -203)
- Databricks Data Engineer Associate